

CS 162

Computer Architecture

University of California, Riverside

Bingyao Li
<http://libingyao.github.io/>

Course Administration



- **Instructor:** Bingyao Li (bingyaol@ucr.edu)
- **Lecture:** MWF 9:00am – 9:50am
- **Location:** WCH 143
- **Logistic:** In person lectures.
No remote participation.

The course administration is subject to change based on the University policies

Course Administration



- **Office hour:** WCH 358
Mon 10:00am– 11:00am
- **TA:** Sixu Tan (sixu.tan@email.ucr.edu)
Office Hours: **WCH 363 Thursday 3:00pm-4:00pm**
- **URL:** Elearn
Piazza: <https://piazza.com/ucr/winter2026/cs162>
- **Text:** *Computer Architecture: A Quantitative Approach; John Hennessy and David Patterson. Sixth Edition - Morgan & Kaufmann.*
Computer Organization and Design - The Hardware/Software Interface By David Patterson and John Hennessy Fifth Edition - Morgan & Kaufmann.
- **Slides:** PDF on Canvas before each lecture
- **ACK:** Profs. Rami Melhem(Pitt), Xulong Tang(Pitt), Mahmut Kandemir (PSU)

Grading Information



- **Grade determinants**

– Midterm exam	20%
– Final exam	20%
– 1 Programming Project	40%
– 4 homeworks	20%
» To be submitted on Canvas by 23:59 (PT) on the due date. No late assignments will be accepted.	

- **In person, in class, close note, synchronized** exams.
- Let the instructor and the TA know about exam conflicts ASAP
- Grades will be posted on Canvas
 - Must submit email request for change of grade after discussions with the instructor
- Attending the class is very important
 - I hope to see all students in the class room.
 - No recording.

Academic Integrity & Other Policies



- Please read the syllabus in Canvas
- Any apparent cases of cheating and collaboration on exams, project, or assignments will be treated as academic dishonesty.

Course Schedule



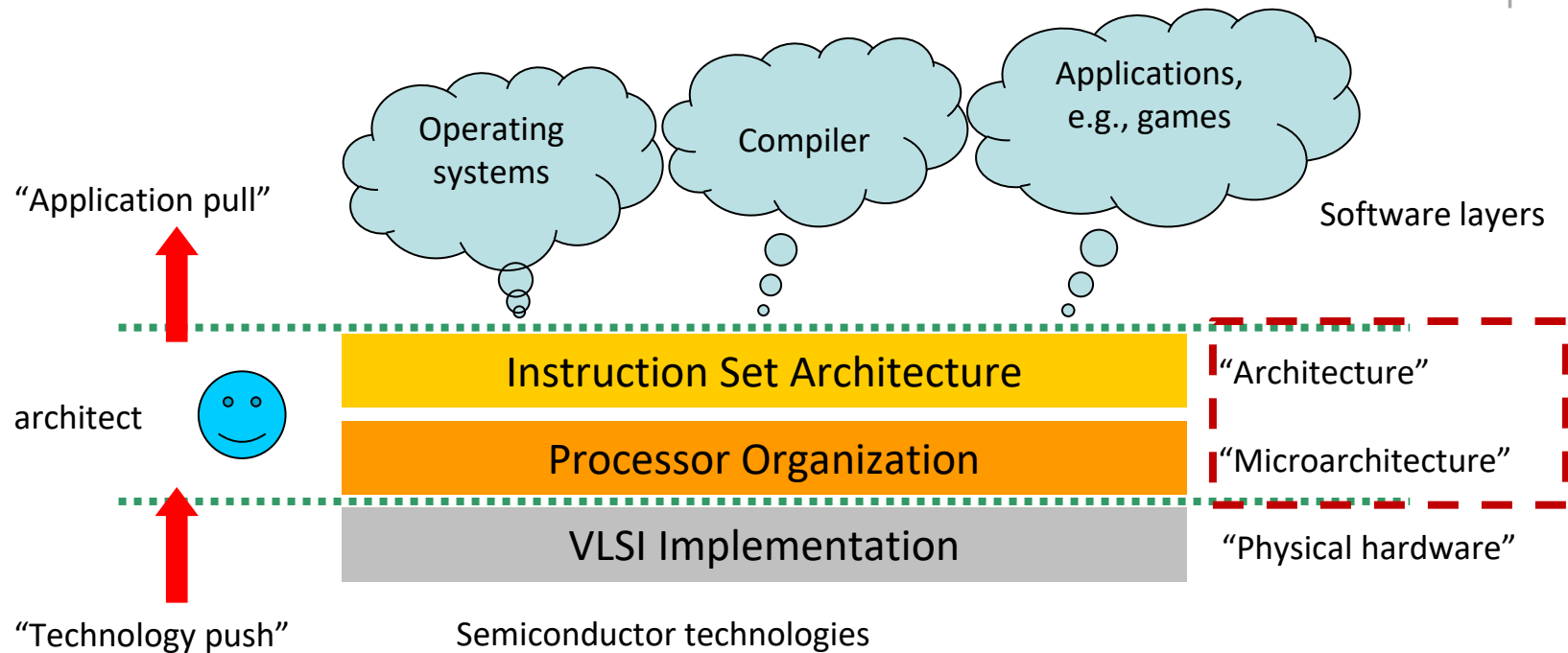
- Based on the basics of computer architecture (CS161)
 - In case you need review the basics:
 - » Computer Organization and Design - The Hardware/Software Interface By David Patterson and John Hennessy Fifth Edition - Morgan & Kaufmann.
- Schedule:
 - 2 weeks: review of the MIPS ISA and pipelined datapath design
 - 2 weeks: superscalar datapath design issues
 - 2 weeks: memory hierarchies and memory design issues
 - 2 weeks: multiprocessor/multicore design issues
 - 2 weeks: throughput processors and GPUs
 - Detailed schedule on syllabus.

Today's Class



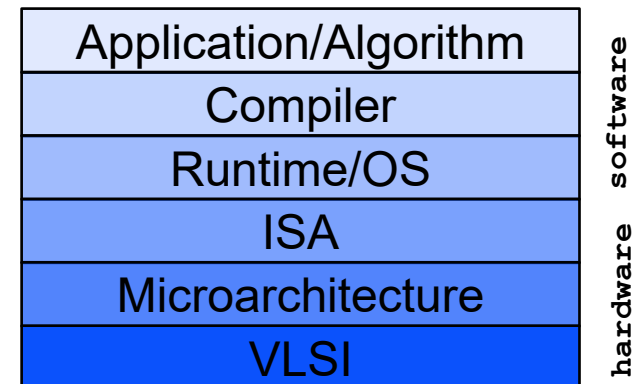
- Lecture
 - What is computer architecture
 - Why we care about it
 - What is the trend (the past and the future)

What is Computer Architecture? (§1.3)



Models of Parallel executions:

- Instruction Level parallelism (ILP)
- Data-level parallelism (DLP)
- Thread-level parallelism (TLP)
- Request-level parallelism (RLP)
- Memory-level parallelism (MLP)

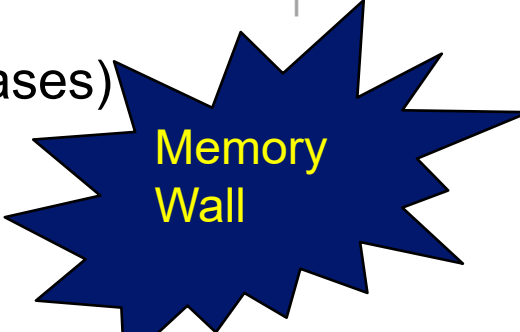


Flynn's Taxonomy

<i>classic von Neumann</i> SISD Single instruction stream Single data stream Single Core	(SIMD) Single instruction stream Multiple data stream GPUs, Vector machines
MISD Multiple instruction stream Single data stream Does it make sense? Yes, systolic array. TPUs	(MIMD) Multiple instruction stream Multiple data stream Multi-core

Trends in Technology (§1.4)

- Integrated circuit technology
 - Transistor density: +35%/year (feature size decreases)
 - Die size: +10-20%/year
 - Integration overall: +40-55%/year
- DRAM capacity: +25-40%/year (growth is slowing)
(memory usage doubles every year)
- Flash capacity: +50-60%/year
 - 8-10X cheaper/bit than DRAM
- Magnetic disk technology: +40%/year
 - 8-10X cheaper/bit than Flash and 200-300X cheaper/bit than DRAM
- Clock rate stopped increasing

A blue starburst graphic with the text 'Memory Wall' in yellow.

Memory Wall

Bandwidth and Latency

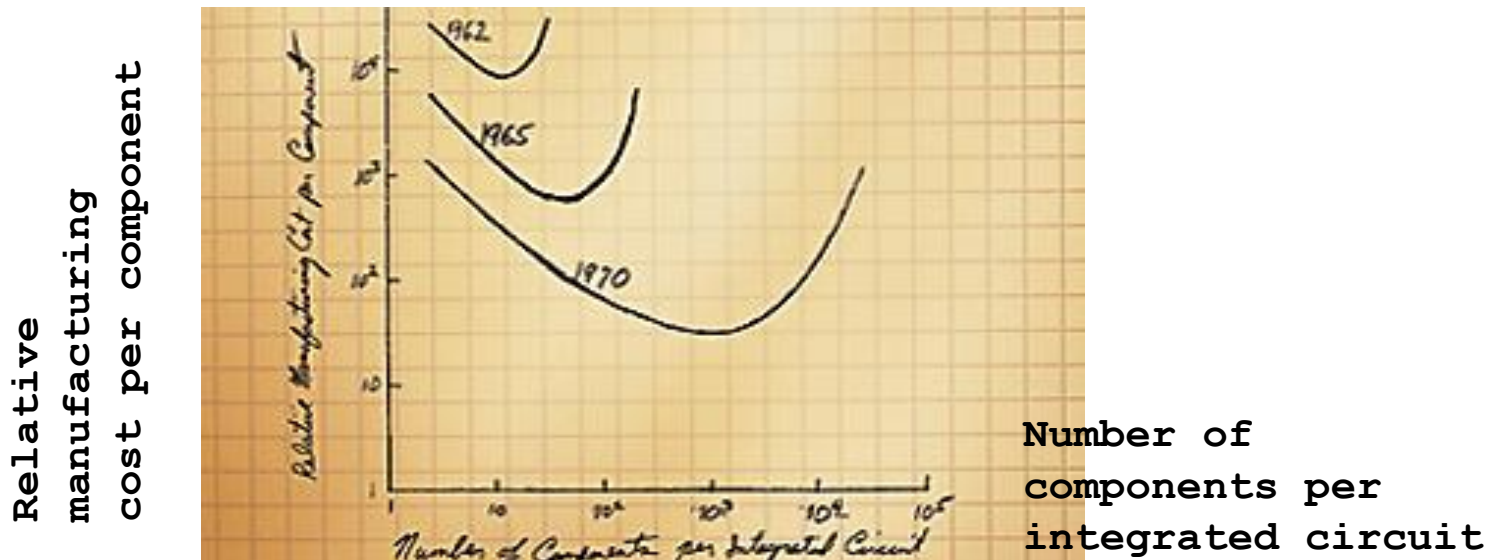


Latency lags bandwidth (in the last 40 years)

- Bandwidth or throughput
 - Total work done in a given time
 - 32,000 - 40,000X improvement for processors
 - 400 - 2400X improvement for memory and disks
- Latency or response time
 - Time between start and completion of an event
 - 50 - 90X improvement for processors
 - 8 - 9X improvement for memory and disks

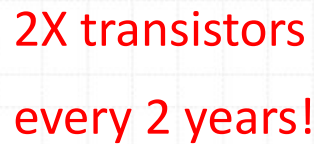
Transistors and wires

- Feature size
 - Minimum size of transistor or wire
 - 10 microns in 1971 to 22 nm in 2012 to 16 nm in 2018 to 7nm and 2020 to 5nm
 - Transistor performance scales linearly
 - Wire delay per unit length does not improve with feature size!
 - Integration density scales quadratically



Moore's note (1965)

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important as other aspects of technological progress – such as processing speed or the price of electronic products – are linked to Moore's law.



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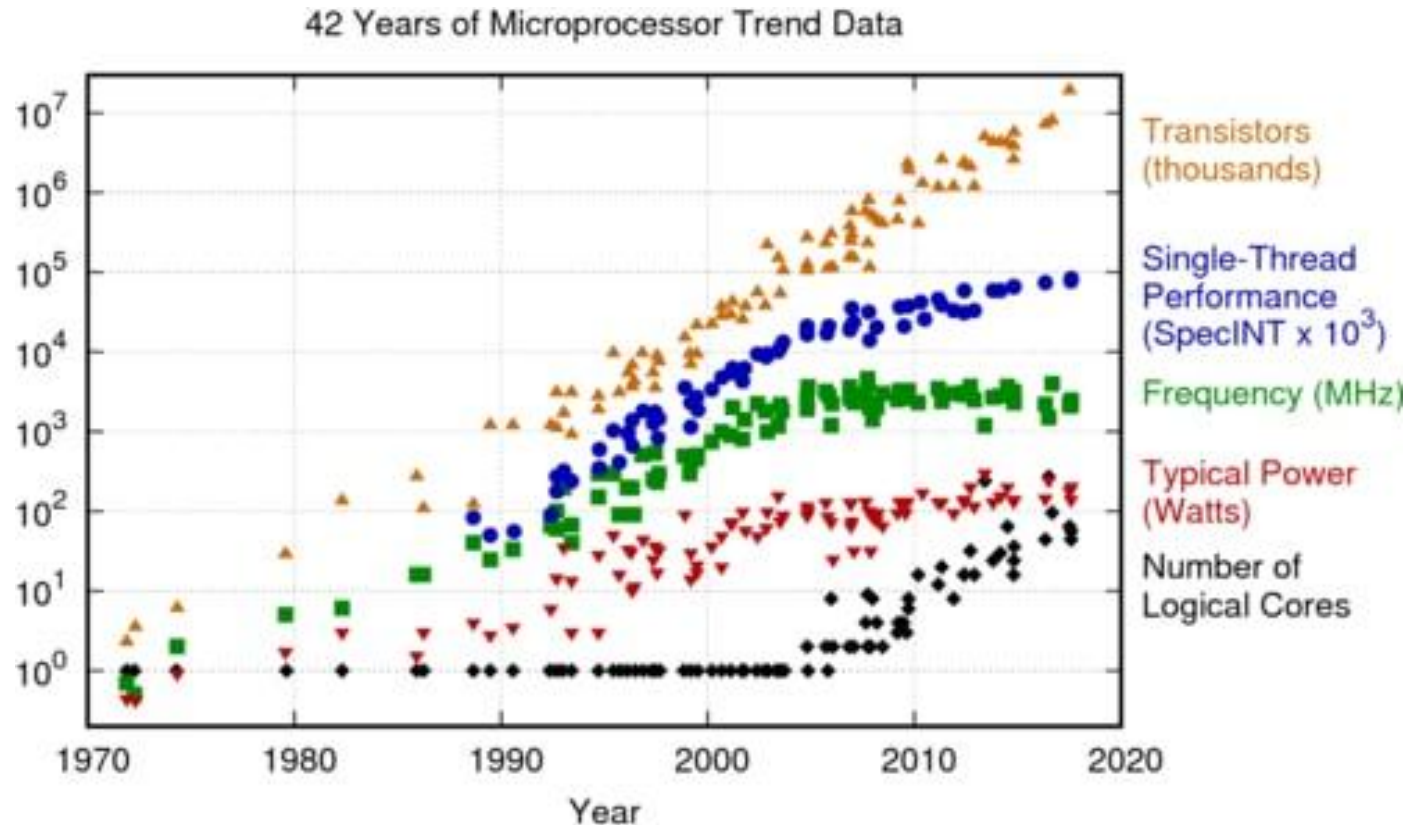
Dennard Scaling



- In 1974, Robert H. Dennard observes that transistor dimensions could be scaled by -30% (0.7x) every technology generation.
 - Their area reduces by 50%.
 - To keep the electric field constant, the voltage, V , is reduced by 30% (0.7x).
 - Circuit delays reduce by 30% (0.7x).
- Therefore
 - The 30% reduction in delay allows an increase in operating frequency, f , by about 40% (1.4x).
 - The 30% reduction in all distances and related 50% drop in areas lead to a decrease in capacitance, C , by 30%.
 - As a result, power consumption in turn decreases by 50%.
- In a nutshell, in every technology generation, if the **transistor density doubles (Moore's Law)**, the circuit becomes 40% faster, and power consumption (with twice the number of transistors) **stays** the same.

$$\text{Power} = \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency}$$

End of Moore's Law (and Dennard Scaling)

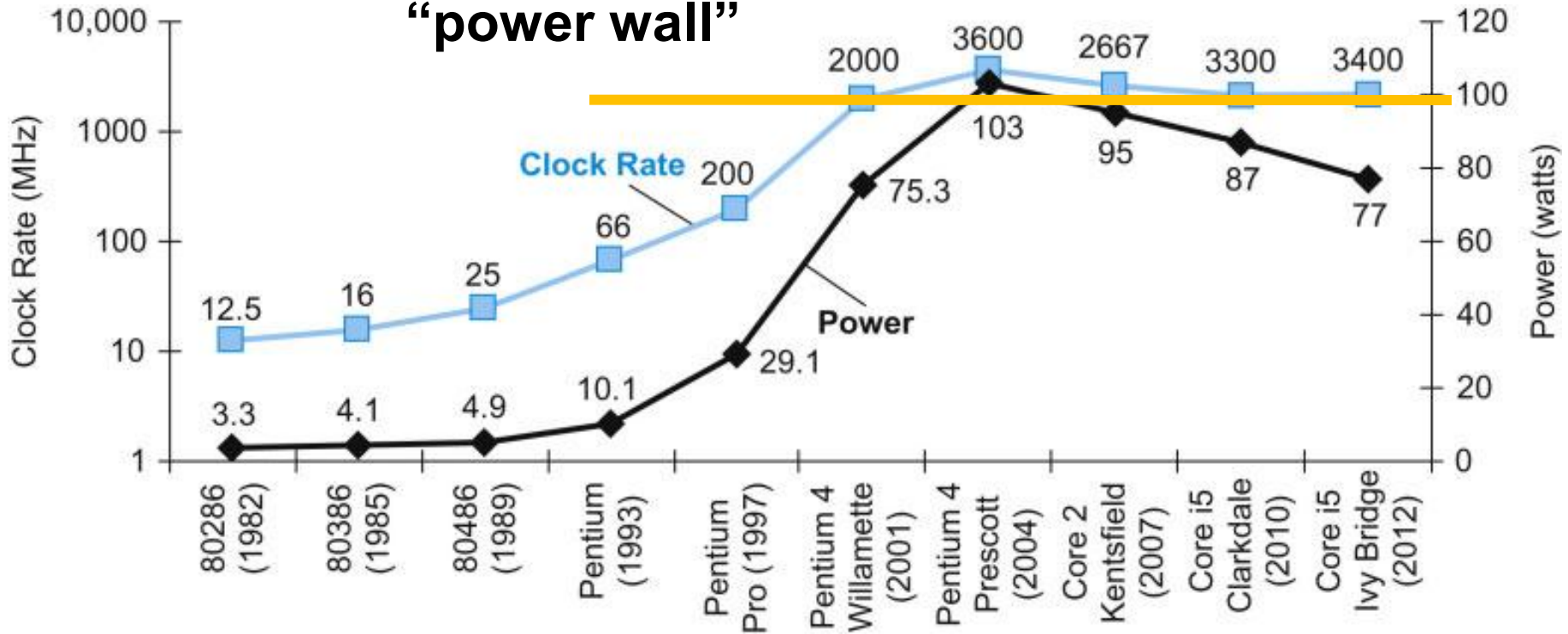


Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten
New plot and data collected for 2010-2017 by K. Rupp

But What Happened to Clock Rates and Why?



**Clock rates hit a
“power wall”**



- In CMOS IC technology

$$\text{Power} = \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency}$$

The Move to Multicore Processors

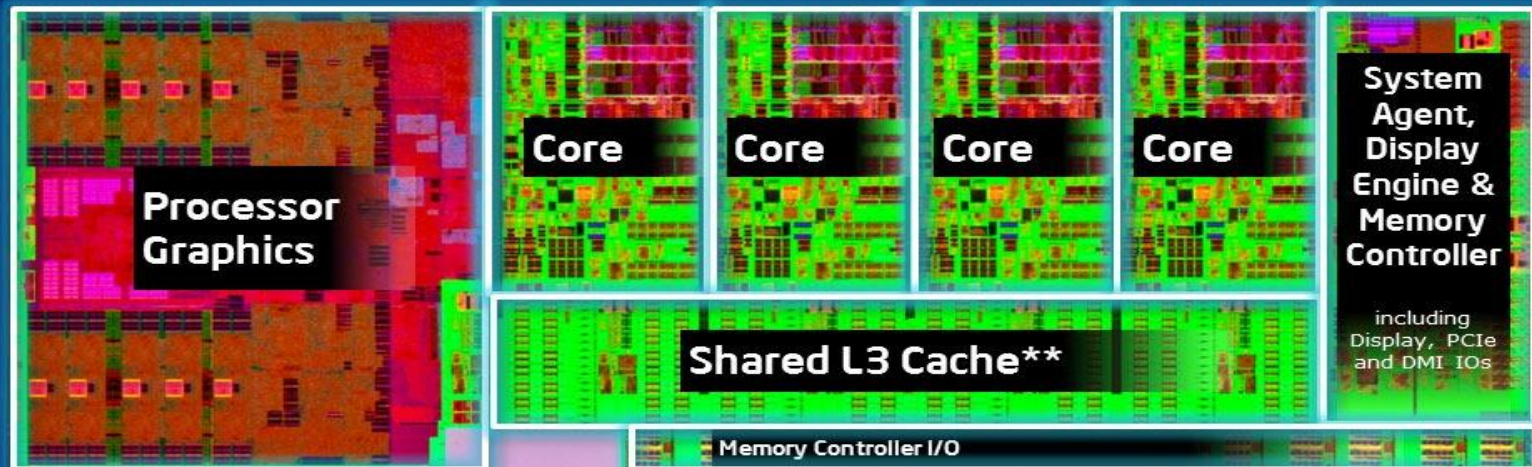
- The power challenge has forced a change in the design of microprocessors
- As of 2006 all server companies were shipping microprocessors with multiple cores per chip (processor)

Product	AMD Opteron X	Intel i7 Haswell	IBM Power 7+
Release date	2013	2013	2012
Technology	28nm bulk	22nm FFET	32nm SOI
Cores/Clock	4/2.0 GHz	4/3.5GHz	8/4.4 GHz
Power (TDP)	22W	84W	~120 W

- Plan of record was to double the number of cores per chip per generation (about every two years)

Intel Haswell

4th Generation Intel® Core™ Processor Die Map *22nm Haswell Tri-Gate 3-D Transistors*



Quad core die shown above | Transistor count: 1.4Billion | Die size: 177mm²

** Cache is shared across all 4 cores and processor graphics

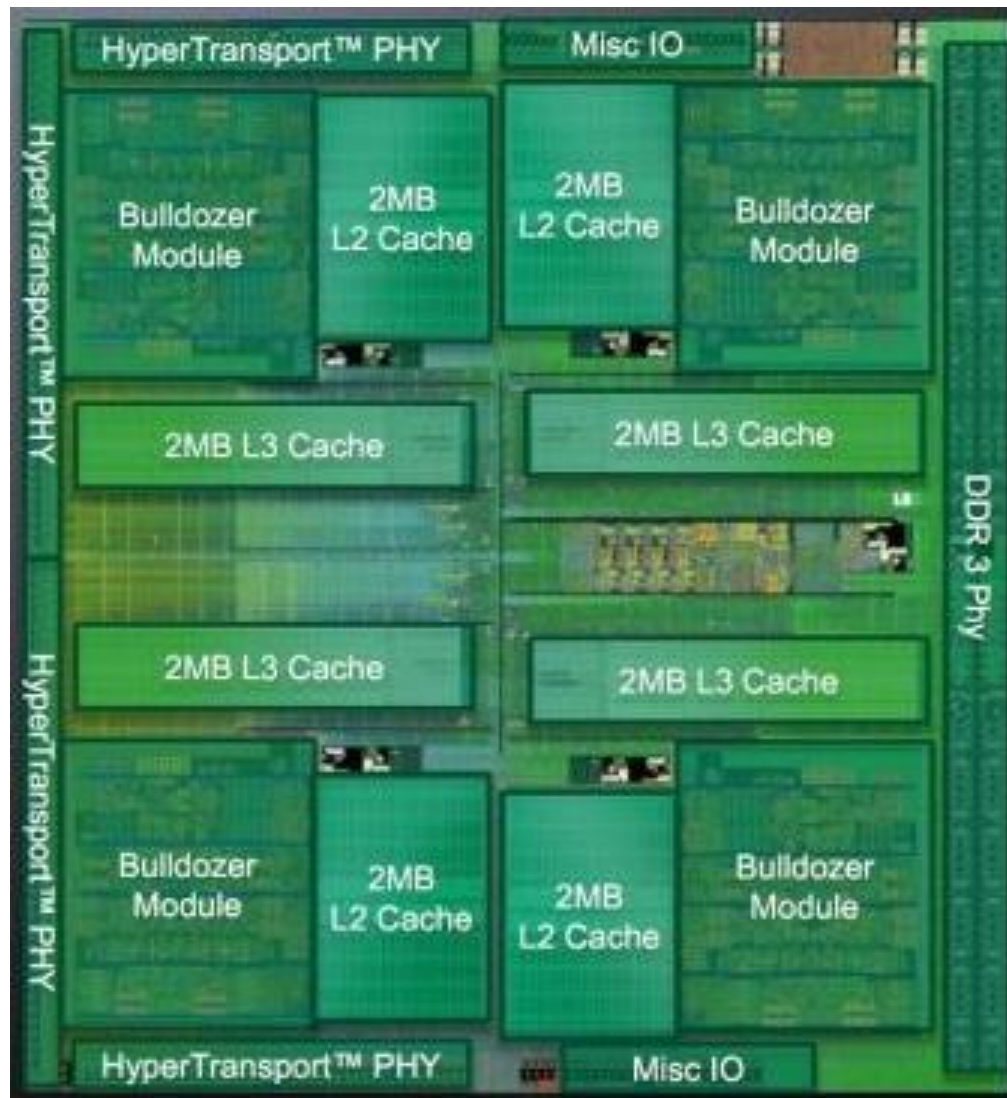
All products, dates, and figures specified are preliminary based on current expectations, and are subject to change without notice.

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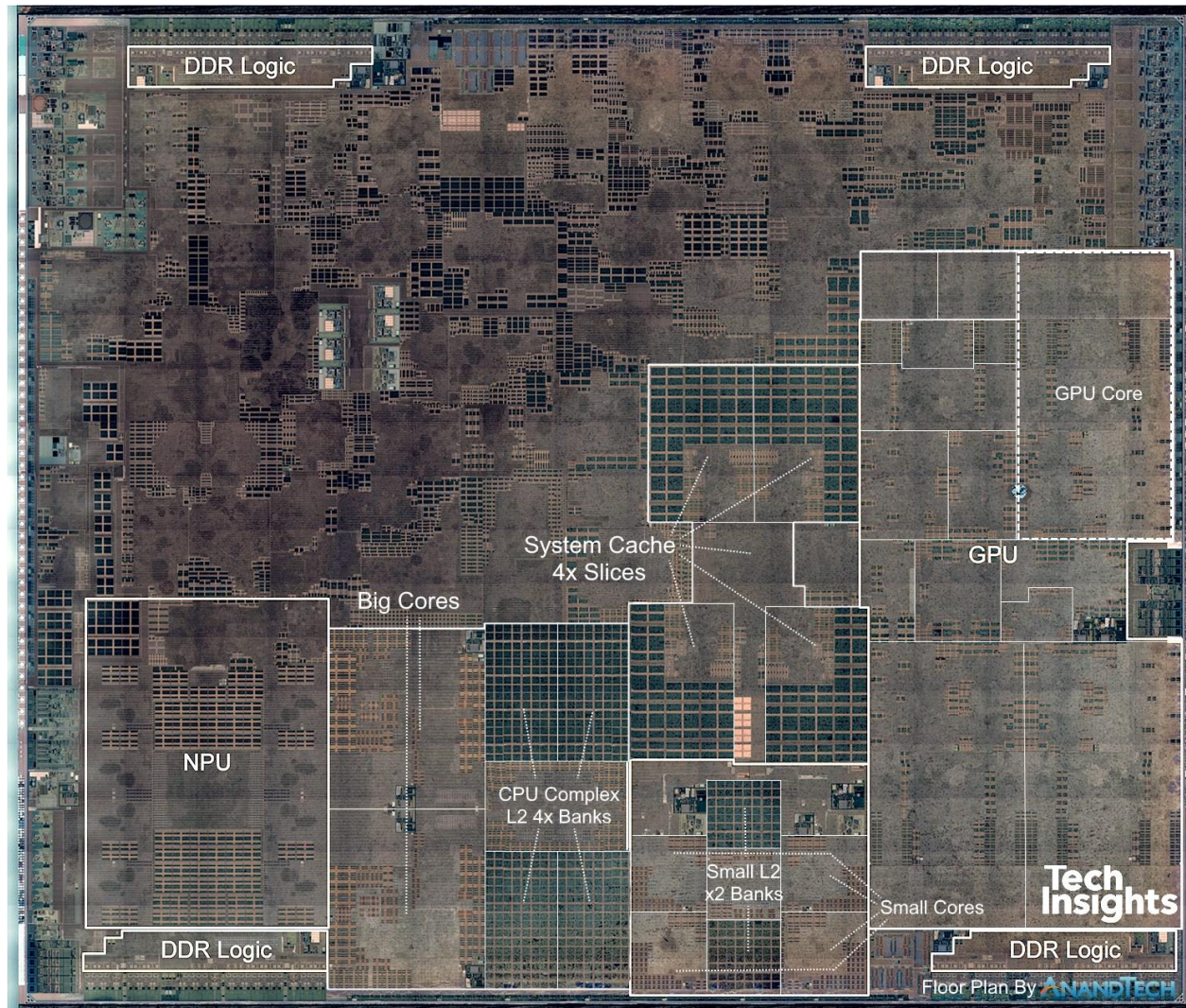
AMD Opteron (Bulldozer)



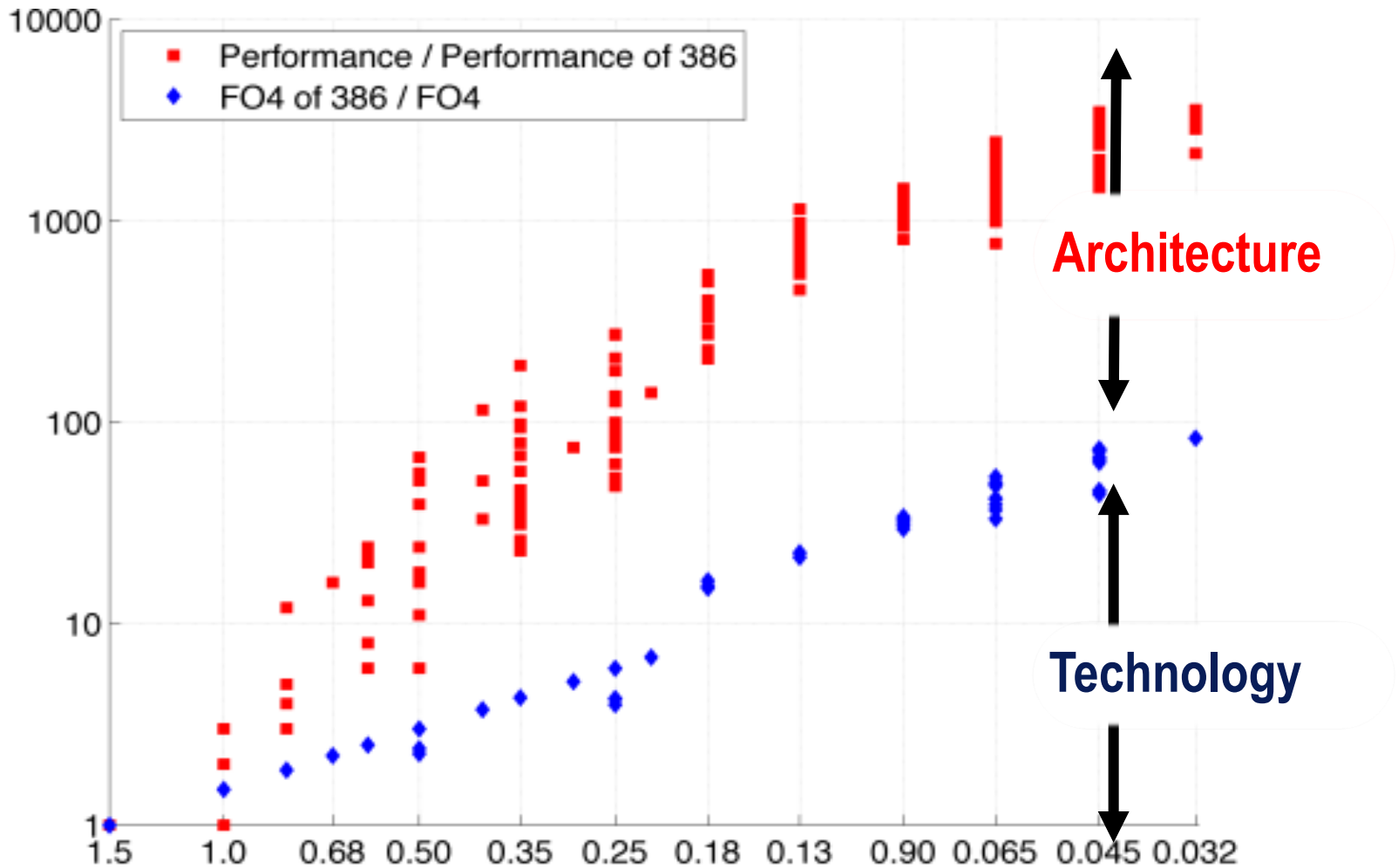
Apple A6 (iPhone 5)



Apple A12



How much benefit from whom?



Danowitz et al., CACM 04/2012, Figure 1

The Trend

- End of Moore's Law and Dennard Scaling since 2004
 - Memory wall
 - » The growing disparity of speed between CPU and memory outside the CPU chip
 - Power wall

Computer architecture designs and innovations are the key to continue providing the application performances and speedups.

- Simpler core designs
 - Expensive inter-core communication and coordination
- Heterogeneous multi-core processors
 - NPU, Tensor core, Ray tracing core, etc
- Domain specific processors
 - AI and computer vision
 - Autonomous driving

